

Point 1

The Orbit Determination process to obtain the estimate of the six state vector's components and the associated uncertainties is provided by the Matlab script *mysolution*.

The script is divided in sections to make more clear all the passages:

- **Load the observables** from the *Observables.txt* file and put them in three vectors of epochs, range and range rate data, where each line is the value at the correspondent epoch;
- **Check the observables** data of range and range rate by plotting them separately in figure 1 as the time of the observations varies;

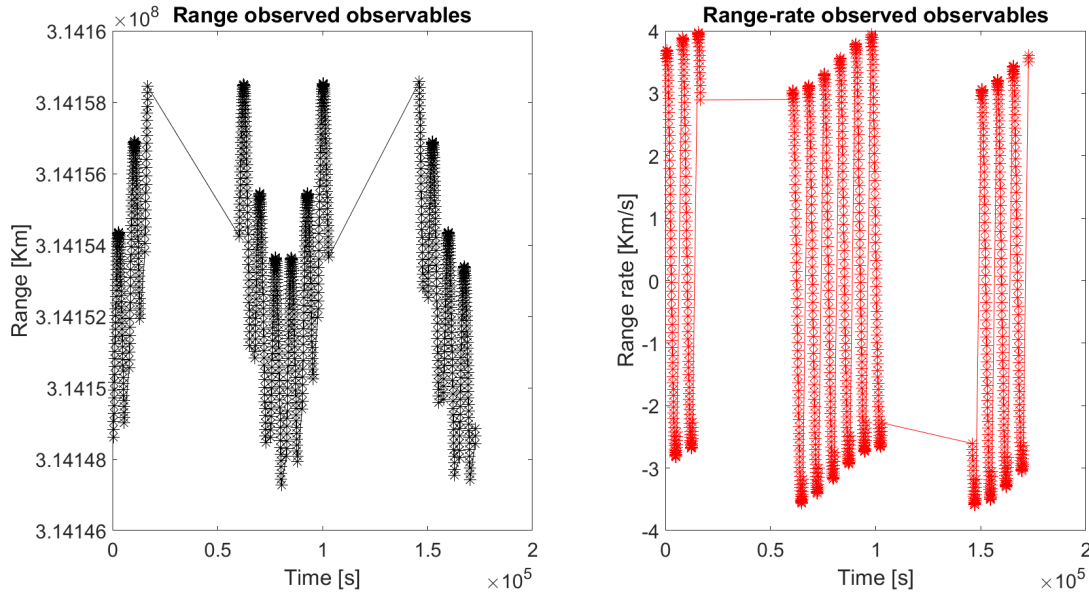


Figure 1: Observables data at each time

- **Set the initial conditions** given at time $t_0=0$ and with respect to Mars. In particular they are set: the *a priori state vector* with the given values of position, velocity, gravitational parameter and atmospheric density at the initial altitude; the general data and the uncertainties of the *a priori state vector*'s components, which are suddenly used to define the diagonal *a priori covariance matrix*.
- **Weighted least-squares solution with a priori information**

The main part of the script is the fourth section with the passages to build the filter for the least-squares solution. After setting all the initial values we built the weighted

State vector	Value	Uncertainty
X0	2505.91 Km	0.00626 m
Y0	2505.91 Km	0.00701 m
U0	-2.53562 Km/s	6 μm
V0	2.61708 Km/s	6 μm
GM	4.2828 Km^3/s^2	$7 * 10^{-5} Km^3/s^2$
Rho(z0)	0.001 Kg/Km^3	$8 * 10^{-5} Kg/Km^3$

Table 1: State vector and associated uncertainties after 5 iterations.

matrix as a diagonal matrix with alternated elements relative to range and range rate on the diagonal.

Then the loop on number on iterations (5) starts with the integration of the state and the state transition matrix for all the given observation's time.

The internal loop on the number of observables allow us to compute the H matrix and all the computed range and range-rate.

Note: in this loop there is also the definition of the ground station position and velocity with respect to Mars, which change on time because we consider the rotation of the Earth (in a counterclockwise way with respect to Mars). In particular the ground moves with a circular motion that starts from an initial angle of 200° with a rotation's velocity equal to that of the Earth.

After closing the internal loop we computed the residuals and built the residuals matrix as the same as the weighted matrix. Then there is the state correction with the formula for the case of *a priori information*. After this the state is updated and also the deviation from the original a priori state.

Note on Functions: In *Model and transition* we have to pay particular attention on the definition of the equations of motion showed in the equation 1, where we have to add the drag acceleration, also in the Dynamical Model where in the terms related to the accelerations appears the drag acceleration due to the Mars' atmosphere. Consequently also in the function where it is defined the A matrix we have to compute all the derivatives in the correct way taking into account all the state vector's components.

$$\dot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_{drag} \quad (1)$$

Results

In table 1 there are shown the values of the state vector and the associated uncertainties after 5 iterations. It is possible to notice that these values are very close to those of the state vector given a priori therefore the used method appears to be correct and capable of determining values with uncertainties even smaller than the starting ones.

In fig. 2 and in fig. 3 there are the residuals of the range and range-rate for the first iteration.

RANGE	MEAN	RMS	SOS	RATE	MEAN	RMS	SOS
iter0	3.02	19.77	1.563×10^{12}	ter0	7.28×10^{-3}	1.233×10^{-2}	9.495×10^{13}
iter1	-2.48×10^{-1}	0.5038	1.014×10^9	iter1	4.71×10^{-5}	5.597×10^{-4}	1.956×10^{11}
iter2	-1.00×10^{-1}	0.1563	9.758×10^7	iter2	-4.38×10^{-5}	8.353×10^{-5}	4.356×10^9
iter3	-9.57×10^{-3}	0.01099	4.824×10^5	iter3	-2.02×10^{-6}	4.753×10^{-6}	1.410×10^7
iter4	-3.10×10^{-4}	3.792×10^{-4}	5.745×10^2	iter4	-1.03×10^{-7}	3.550×10^{-7}	7.867×10^4
iter5	2.86×10^{-5}	1.095×10^{-4}	4.788×10	iter5	-7.82×10^{-9}	3.979×10^{-8}	9.884×10^2

Table 2: MEAN,RMS,SOS of range and range rate for all the iterations

It is evident that the distribution has a particular evolution, it means that there is a particular correlation between the computed and observed observables. In fig. 4 and fig. 5 there are the plots of all the range and range-rate residuals for each observation after 5 iterations. Looking at the distribution for both the plots it's possible to see that the points are clearly spread in a visible limit close to zero (white noise), which means that the difference between our dynamic model and the measurement model is minimal.

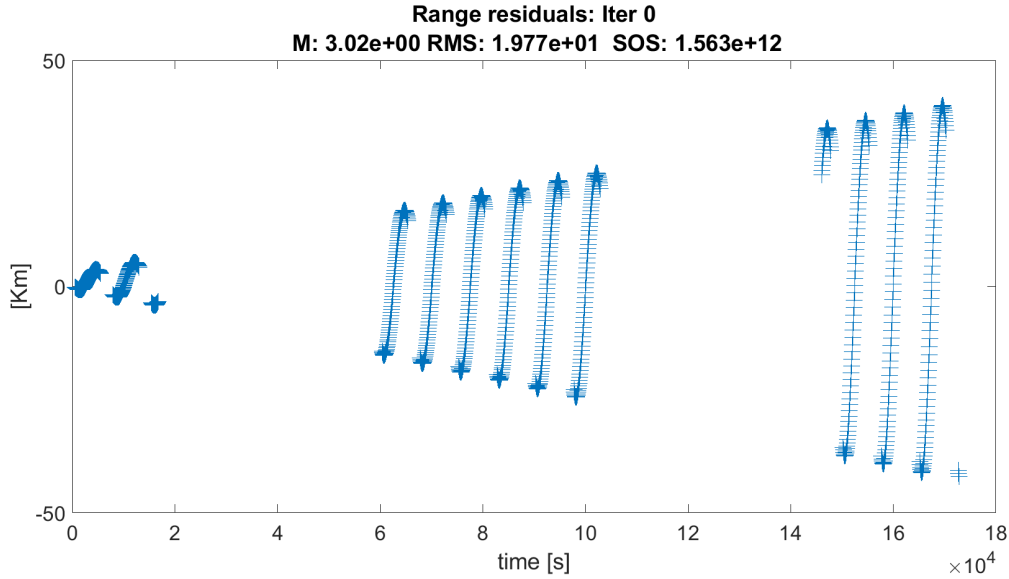


Figure 2: Range residuals at iter 0

But to check the goodness of the solution it is necessary to compute the SOS, MEAN and RMS, which are reported in table 2 for all the iterations. This table shows clearly the convergence of the procedure because from top to the bottom all the parameters decrease until they reach better values. In fact we can notice that after the last iteration: both the range types' MEAN is very close to zero, it clarifies what was explained before about the distribution of the residuals; the RMS, which has to be equal to the accuracy of the ground station's range and range-rate measurements, is very close to the correct value for the range-

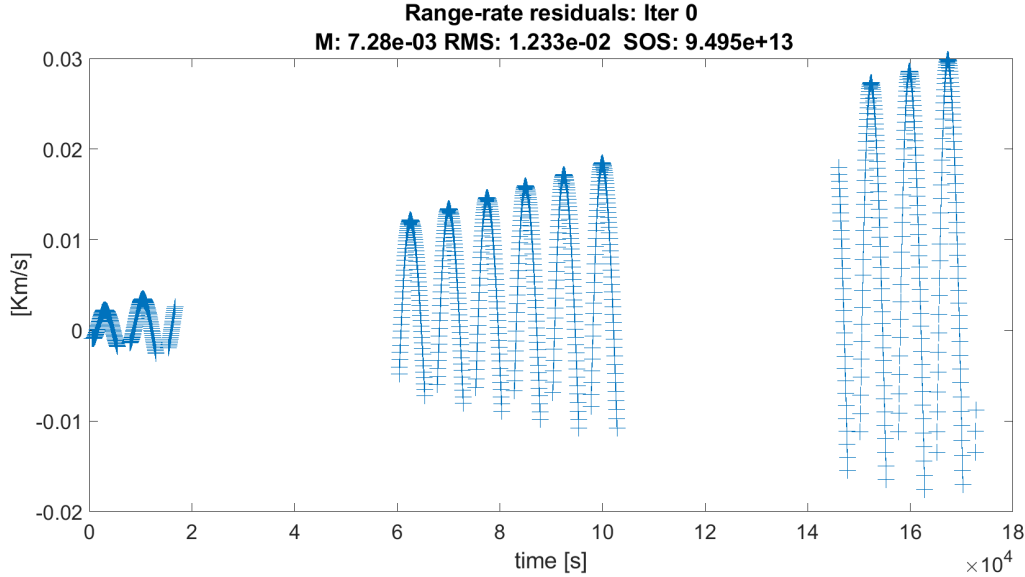


Figure 3: Range-rate residuals at iter 0

Range type	MEAN	RMS	SOS
Range	$3.09 * 10^{-5}$	$1.089 * 10^{-4}$	999
Range rate	$6.70 * 10^{-9}$	$3.643 * 10^{-8}$	999.4

Table 3: MEAN, RMS, SOS for each type of range at the last iteration with new weights.

rate and very far from the expected value for the range; the SOS reflects the behaviour of the RMS, it should be equal to the number of observables (999) but only the range-rate approaches the correct value.

After this discussion is clear that we can try to change the uncertainties of the range-rate and range measurements and the correspondent weighted matrix to obtain a good values for the previous parameters. In particular we have to obtain a greather SOS so we decrease the *sigma-obs-range* and the *sigma-obs-range-rate*. After some computations the better values for the *sigma-obs-range* is $1.089 * 10^{-4}$ and for the *sigma-obs-range-rate* $3.642 * 10^{-8}$. In the tab. 3 it is possible to check the final parameters which allow us to say that we have reached a qualitatively good solution. In table 4 there is the final state vector and uncertainties and in fig. 6 and fig. 7 the evolution of the residuals.

Point 2

It is possible to determine which of the two types of observables is most valuable for the determination of the probe state by looking at the statistical parameters of the residual's distribution at the end of the process described in the solution of point 1. In particular we have to look at the RMS in table 3; it represents the square root of the means of the squares

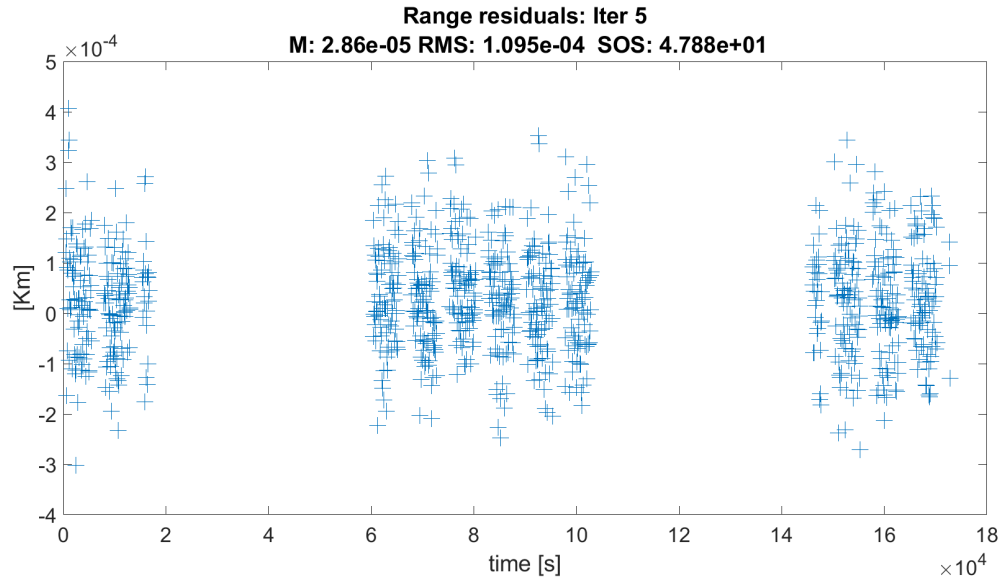


Figure 4: Range residuals at iter 5

State vector	Value	Uncertainty
X0	2505.91 Km	0.00491 m
Y0	2505.91 Km	0.00579 m
U0	-2.53562 Km/s	4.5 μm
V0	2.61708 Km/s	4.8 μm
GM	4.2828 Km ³ /s ²	5.6 * 10 ⁻⁵ Km ³ /s ²
Rho(z0)	0.001 Kg/Km ³	6.4 * 10 ⁻⁵ Kg/Km ³

Table 4: State vector and associated uncertainties after 5 iterations and better weights.

of the deviation values which in our case corresponds to the error associated to the difference between the computed and observed observables. This means that the lower is the RMS value, the greater is the weight attributed to that type of observable for our measurements. This is the principle on which the weighted least-squares solution is based, in fact thanks to this trick it is possible to attribute more importance for better observations.

Results

It's evident that from table 3 the lower RMS is the one attributed to the range-rate, it differs from the range's RMS by four order of magnitude meaning that the most valuable observable for the determination of the probe state is the range-rate.

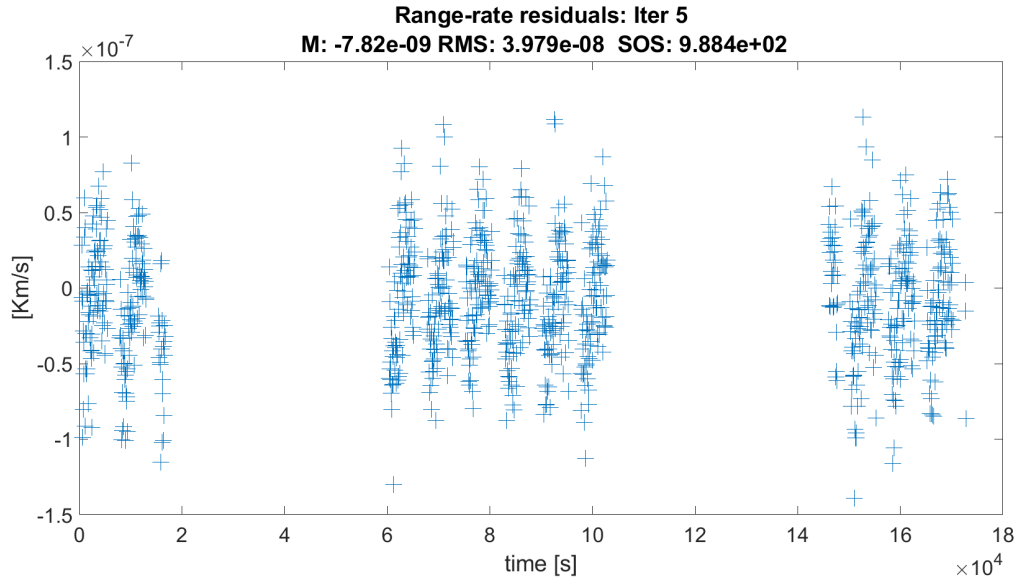


Figure 5: Range-rate residuals at iter 5

Point 3

To obtain the evolution of the altitude and its uncertainty from t_0 to t_0+3 days we have to propagate the covariance matrix. As the first point we iterate the state and state transition matrix and propagate the covariance matrix for all the needed times. Then we express the altitude as a function of the position of the spacecraft and the associated uncertainty using the given expression for the uncertainty of derived quantities.

Results

In figure 8 it is plotted the altitude of the spacecraft in the given period of time. The evolution has a sinusoidal behaviour with minimum value of 151 Km and a maximum value of 936 Km. In figure 9 it is plotted the altitude's uncertainty of the spacecraft.

Point 4

In the script named *mysolution_pt4* there is the computation of the new results considering the drag coefficient as one of the a priori parameters. From the point of view of the procedure it's exactly the same of the first point changing the size of the a priori state vector paying attention to the A matrix and its relative derivative with respect to the new parameter.

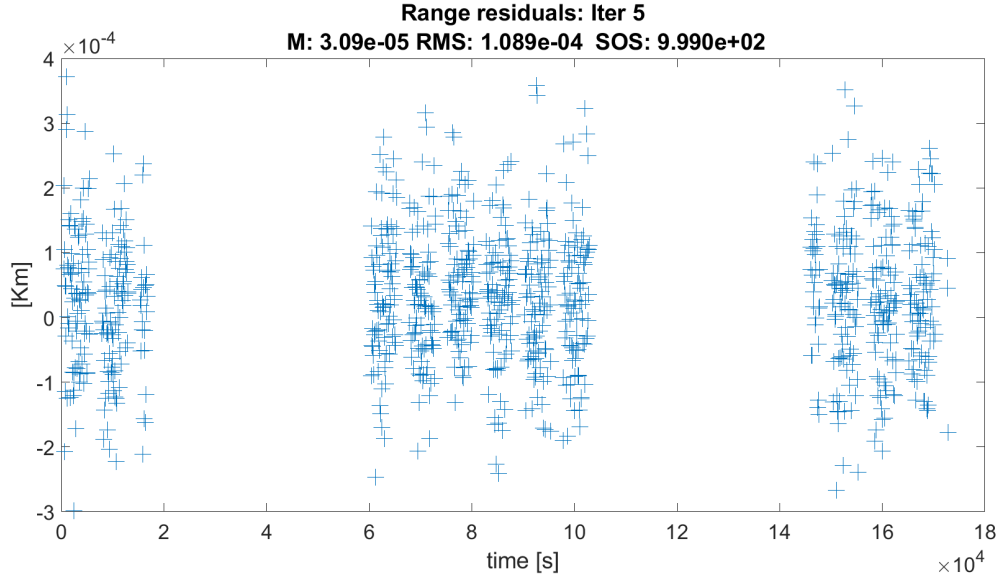


Figure 6: Range residuals at iter 5 after changing weights

Results

In figure 10 and 11 there are the results in terms of residuals at the last iteration.

In table 5 there are the state vector components after 5 iterations. We can see that all the parameters that not depend on the drag coefficient remains the same of the point 4 and the density at initial altitude changes with an order of thousandths. It's interesting to notice that after all the iterations the drag coefficient remains exactly the same as the a priori values both in terms of value and uncertainty even if, looking step by step inside the previous iterations, it changes but with a very low order. Furthermore the value of the density at the initial altitude changes from 0.001 Kg/Km^3 to 0.002 Kg/Km^3 , it is a direct consequence of the propagation of the drag coefficient even if the variation is lower than the significant figures given at the a priori value.

Note: in this point we considered the optimal weights of the last step of point 1, which return acceptable value of MOS, RMS and SOS.

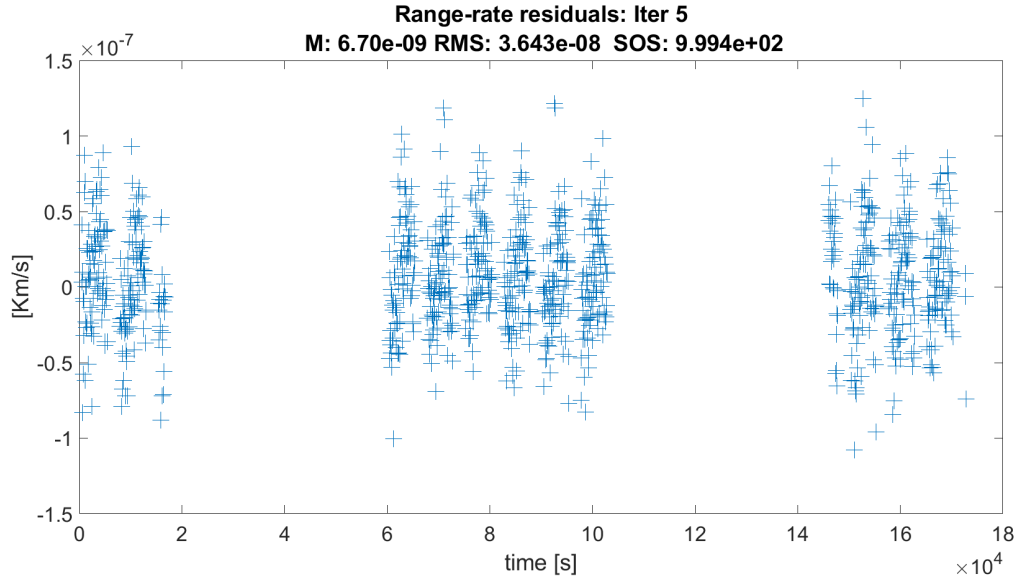


Figure 7: Range-rate residuals at iter 5 after changing weights

State vector	Value	Uncertainty
X0	2505.91 Km	0.00491 m
Y0	2505.91 Km	0.00579 m
U0	-2.53562 Km/s	4.5 μm
V0	2.61708 Km/s	4.8 μm
GM	4.2828 Km^3/s^2	$5.6 * 10^{-5} Km^3/s^2$
Rho(z0)	0.002 Kg/Km^3	0.001 Kg/Km^3
Cd	1.00	0.5

Table 5: State vector and associated uncertainties after 5 iterations considering Cd as a state component.

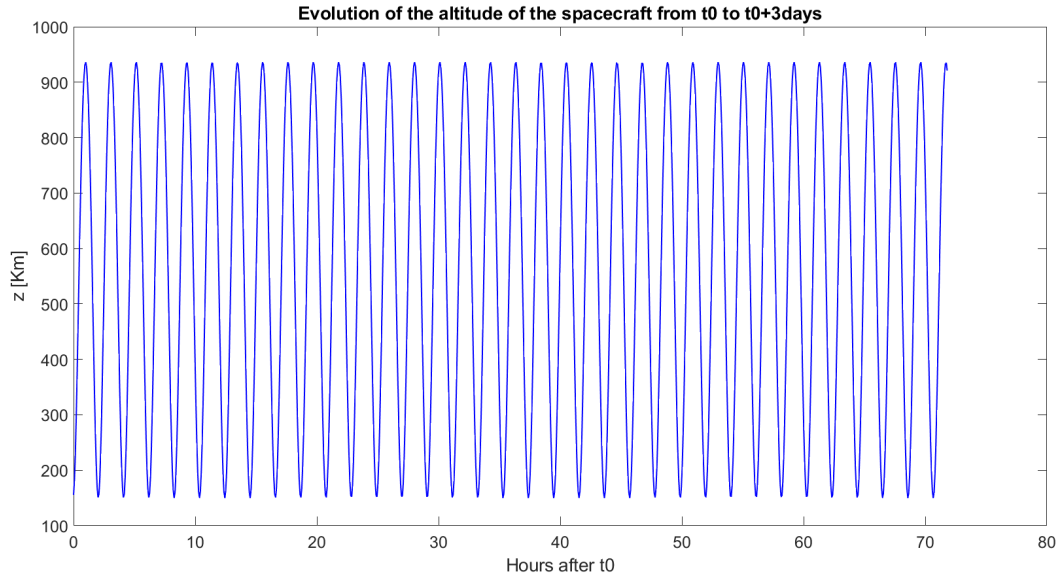


Figure 8: Evolution of the altitude of the spacecraft from t0 to t0+3days

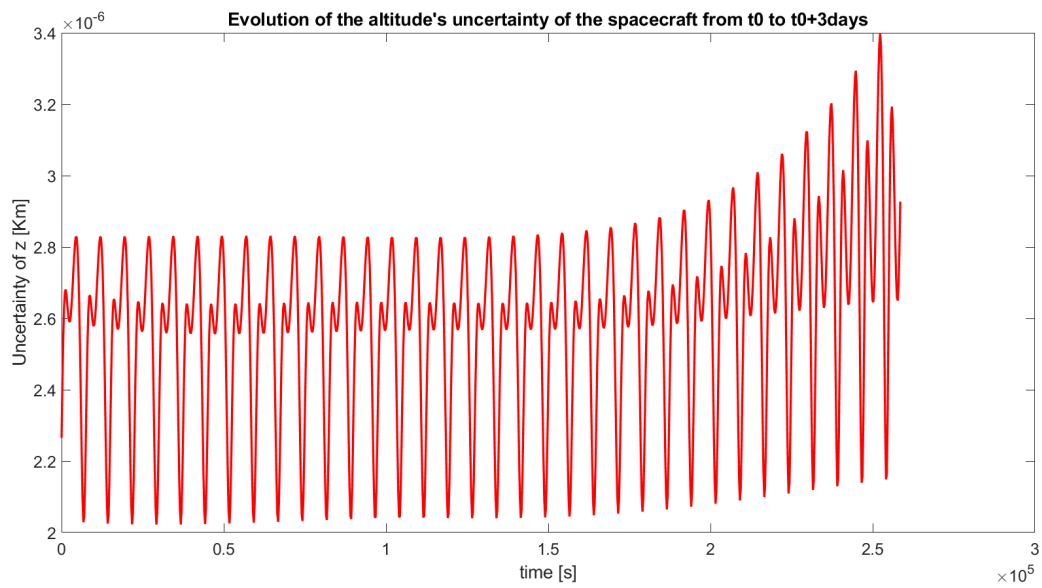


Figure 9: Evolution of the Altitude's uncertainty of the spacecraft from t0 to t0+3days

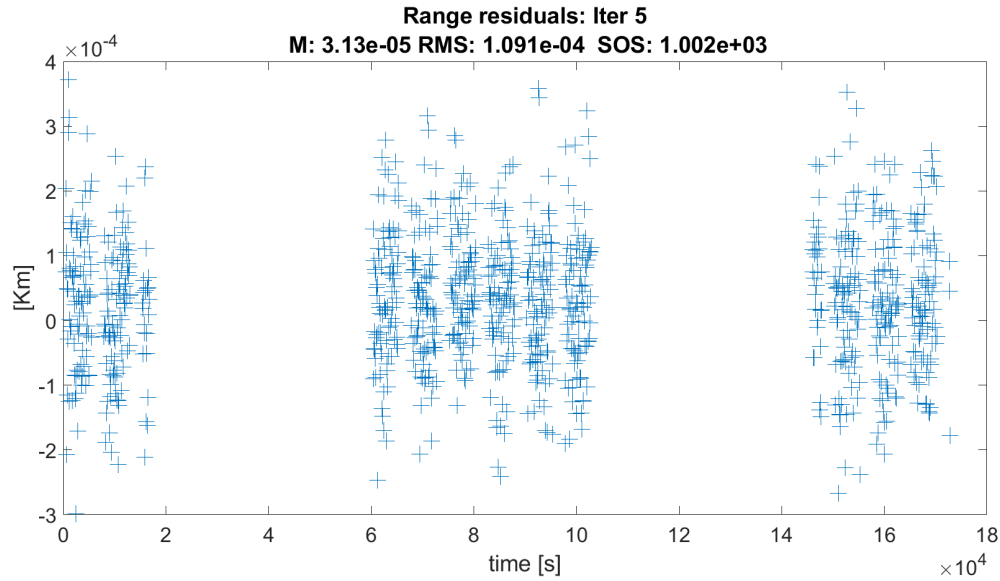


Figure 10: Range residuals at iter 5

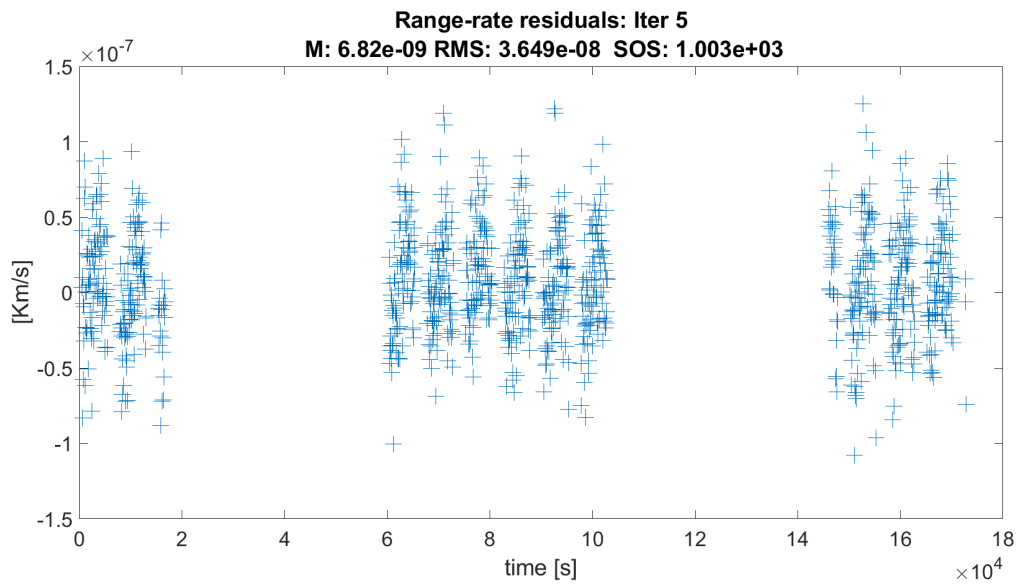


Figure 11: Range-rate residuals at iter 5